

# 01 PROBABILITY FOUNDATIONS (lecture 2, notes p012-p050)

## 1. What statistics is (p012-p016) - not examined, but the framing

statistics = the science of COLLECTION, PRESENTATION, ANALYSIS, INTERPRETATION of numerical data

"All statistics are numerical statements, but not all numerical statements are statistics."

Ram has 100 Rs in his pocket -> NOT statistics (one fact, no data set)  
avg age of 5th class is 10 yrs -> IS statistics (a derived numerical statement)

Four limitations the slides list (worth a one-line recall, low exam weight): 1. not suited to qualitative phenomena 2. does not study individuals 3. statistical laws are not exact 4. liable to be misused (the customer-satisfaction survey only of loyal customers)

## 2. Probability, the intuition (p018-p019)

$P$  near 1 = event extremely likely

$P = 0.5$  = a coin-flip situation

$P$  near 0 = rare event

$0 \leq P(E) \leq 1$  ALWAYS

|            |           |         |
|------------|-----------|---------|
| 0          | 0.5       | 1       |
| impossible | even      | certain |
|            |           |         |
| rare       | coin flip | common  |

## 3. Random experiment (p023)

DEFINITION

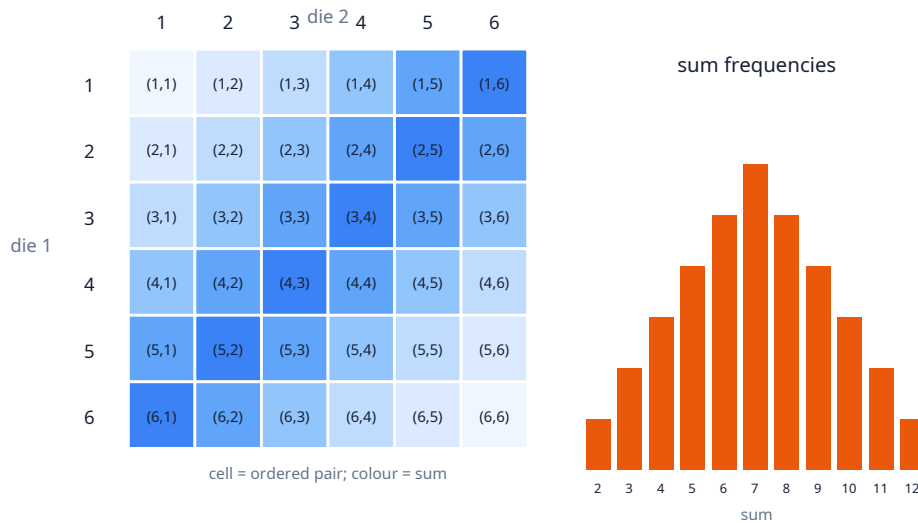
an experiment whose outcome cannot be predicted with certainty in advance, although ALL possible outcomes are known

| EXPERIMENT     | POSSIBLE OUTCOMES        |
|----------------|--------------------------|
| toss a coin    | H, T                     |
| roll a die     | 1,2,3,4,5,6              |
| draw a card    | any one of 52            |
| pick a student | any student in the class |

note: we KNOW all outcomes, we do not know WHICH ONE occurs

## 4. Sample space $S$ (p024)

## Two dice: 36 ordered outcomes



The 36 ordered outcomes for two dice

$S$  = set of ALL possible outcomes

|           |                                      |                            |
|-----------|--------------------------------------|----------------------------|
| one coin  | $S = \{H, T\}$                       | 2 outcomes                 |
| one die   | $S = \{1, 2, 3, 4, 5, 6\}$           | 6 outcomes                 |
| two coins | $S = \{HH, HT, TH, TT\}$             | 4 outcomes                 |
| two dice  | $S = \{(1,1), (1,2), \dots, (6,6)\}$ | $6 \times 6 = 36$ outcomes |

THE COUNTING RULE THAT SAVES MARKS  
 two dice are counted as 36 ORDERED pairs, not 21 unordered  
 (1,2) and (2,1) are DIFFERENT outcomes  
 every dice probability ( $1/36$ ,  $2/36$ ,  $3/36$ ...) assumes 36

The 36 ordered-pair positions, with each cell displaying its sum:

|   | 1 | 2 | 3 | 4  | 5  | 6  | <- die 2 |
|---|---|---|---|----|----|----|----------|
| 1 | 2 | 3 | 4 | 5  | 6  | 7  |          |
| 2 | 3 | 4 | 5 | 6  | 7  | 8  |          |
| 3 | 4 | 5 | 6 | 7  | 8  | 9  |          |
| 4 | 5 | 6 | 7 | 8  | 9  | 10 |          |
| 5 | 6 | 7 | 8 | 9  | 10 | 11 |          |
| 6 | 7 | 8 | 9 | 10 | 11 | 12 |          |

cell position = (row die, column die)  
 printed value = the sum; each cell =  $1/36$

die 1

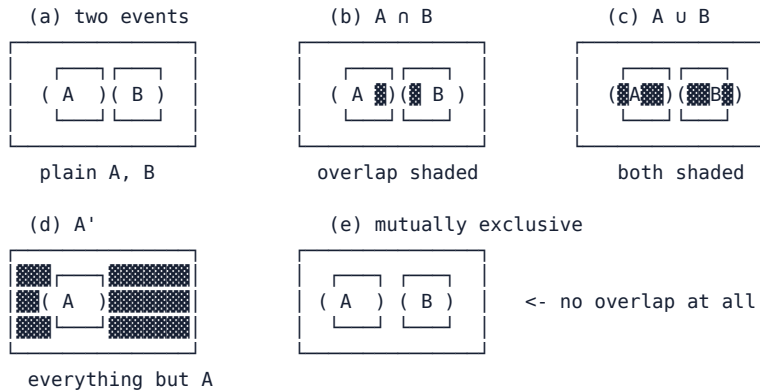
count the 7s: 6 of them  $\rightarrow P(\text{sum}=7) = 6/36 = 1/6$  (the most likely sum)  
 count the 2s: 1 (top-left)  $\rightarrow 1/36$   
 count the 12s: 1 (bottom-right)  $\rightarrow 1/36$   
 the counts run 1,2,3,4,5,6,5,4,3,2,1 (sums 2..12) = the familiar triangle

## 5. Events and set algebra (p030-p040)

|            |                                            |
|------------|--------------------------------------------|
| outcome    | a single possible result                   |
| event      | a SUBSET of $S$ (a collection of outcomes) |
| null event | the empty set, no outcomes                 |

|                                                        |              |           |                          |
|--------------------------------------------------------|--------------|-----------|--------------------------|
| $A'$                                                   | complement   | "not A"   | everything in S not in A |
| $A \cup B$                                             | union        | "A or B"  | in A, or B, or BOTH      |
| $A \cap B$                                             | intersection | "A and B" | in BOTH                  |
| disjoint / mutually exclusive : $A \cap B = \emptyset$ |              |           |                          |

ASCII Venn diagrams:



Demorgan, drawn as the two rules the slides state:

$$(A \cup B)' = A' \cap B' \quad (A \cap B)' = A' \cup B'$$

"neither" = not-A and not-B    "not both" = not-A or not-B

## 6. Mutually exclusive vs independent (p025-p028) - THE classic trap

### MUTUALLY EXCLUSIVE

cannot happen together  
 $A \cap B = \emptyset$   
 $P(A \cup B) = P(A) + P(B)$

### INDEPENDENT

one happening does not change  
the chance of the other  
 $P(A \cap B) = P(A) \times P(B)$   
or  $P(A|B) = P(A)$

THEY ARE NOT THE SAME, AND THEY ARE NOT OPPOSITES

exclusive -> CANNOT both happen -> knowing A KILLS B  
(so NOT independent,  
when  $P > 0$ )

Worked: the die counter-example (p025)

$A = \text{even} = \{2, 4, 6\}$      $B = \text{prime} = \{2, 3, 5\}$   
 $A \cap B = \{2\}$  -> they SHARE the outcome 2  
-> NOT mutually exclusive

$A = \text{even} = \{2, 4, 6\}$      $B = \text{odd} = \{1, 3, 5\}$   
 $A \cap B = \emptyset$  -> mutually exclusive

Worked: coin + die independence (p027)

$A = \text{head}, P(A) = 1/2$   
 $B = 6, P(B) = 1/6$   
 $P(A \cap B) = 1/2 \times 1/6 = 1/12$  -> independent

Worked: draw + replace (p027) -> independent. draw, no replace -> dependent.

Mutual independence of  $n$  events (p028): for EVERY subset of size  $k$  ( $k = 2..n$ ),

$$P(A_{i1} \cap A_{i2} \cap \dots \cap A_{ik}) = P(A_{i1}) P(A_{i2}) \dots P(A_{ik})$$

## 7. Counting: the engine of half the course (p043)

ORDER MATTERS ?

ORDER DOES NOT MATTER ?

**PERMUTATION**

$$P(n,r) = n!/(n-r)!$$

"arrangements"

**COMBINATION**

$$C(n,r) = n!/(r!(n-r)!)$$

"choose", "committee", "sets of"

with REPEATED items:

$$n! / (n_1! n_2! \dots n_k!) \quad n_1, n_2, \dots \text{ are the repeat counts}$$

**WORD TRIGGERS**

"arrange", "lineup", "order", "rank" -> PERMUTATION

"choose", "select", "committee", "set" -> COMBINATION

and  $C(n,r) = C(n,n-r)$  (choosing  $r$  to keep = choosing  $n-r$  to drop)

## 8. The axioms and rules (p044-p045)

**AXIOMS**

$$P(S) = 1$$

$$P(A) \geq 0$$

$$\text{disjoint } A_1, A_2, \dots : P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

**RULES**

$$\text{addition} \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

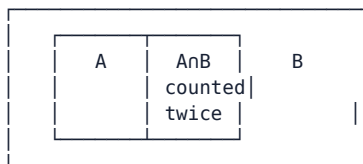
$$\text{disjoint case} \quad P(A \cup B) = P(A) + P(B)$$

$$\text{multiplication} \quad P(A \cap B) = P(A) P(B|A) = P(B) P(A|B)$$

$$\text{independent} \quad P(A \cap B) = P(A) P(B)$$

$$\text{complement} \quad P(A') = 1 - P(A)$$

The addition-rule picture (why you subtract the overlap):



$P(A)+P(B)$  counts  $AnB$  TWICE  
subtract it ONCE

## 9. Worked example 1: tournament (p041-p042)

Four universities 1,2,3,4. Round 1: 1v2, 3v4. Winners play for the title, losers play too.

one outcome is written as a 4-letter code, e.g. 1324:

1 beats 2, 3 beats 4 (round 1) then 1 beats 3, 2 beats 4

$S = \{ 1324, 1342, 1423, 1432, 2314, 2341, 2413, 2431, 3124, 3142, 4123, 4132, 3214, 3241, 4213, 4231 \}$  -> 16 outcomes

(b)  $A = 1$  wins the tournament -> 1 must be FIRST in the code  
 $A = \{1324, 1342, 1423, 1432\}$  4 outcomes

(c)  $B = 2$  gets into the final -> 2 must be 1st or 2nd  
 $B = \{2314, 2341, 2413, 2431, 3214, 3241, 4213, 4231\}$  8 outcomes

(d)  $A \cup B = 12$  outcomes  
 $A \cap B = \emptyset$  because 1 and 2 CANNOT both reach the final  
 $A' = S - A = 12$  outcomes

## 10. Worked example 2: faculty laptops (p046-p047)

6 faculty, 2 chose laptop, 4 chose desktop. Only 2 setups per day, the 2 are chosen randomly.

total outcomes =  $C(6,2) = 15$  (choose 2 of the 6 computers)

(a) both laptops =  $C(2,2)/15 = 1/15$

(b) both desktops =  $C(4,2)/15 = 6/15 = 2/5$

(c) at least one desktop : complement is "both laptops" = A  
 $P(C) = 1 - P(A) = 1 - 1/15 = 14/15$

(d) at least one of each type:

$$1 - P(A \cup B) = 1 - 1/15 - 2/5 = 1 - 1/15 - 6/15 = 8/15$$

(A and B are disjoint, so no overlap to subtract)

## 11. Worked example 3: basketball lineup (p048-p049)

STARTING LINEUP = 2 guards + 2 forwards + 1 center

Part (a): 3 centers, 4 guards, 4 forwards, 1 swing player X

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| no X                | $C(4,2) \times C(4,2) \times C(3,1) = 6 \times 6 \times 3 = 108$ |
| X as guard          | $C(4,1) \times C(4,2) \times C(3,1) = 4 \times 6 \times 3 = 72$  |
| X as forward        | $C(4,2) \times C(4,1) \times C(3,1) = 6 \times 4 \times 3 = 72$  |
| TOTAL = 252 lineups |                                                                  |

Part (b): 5 guards, 5 forwards, 3 centers, 2 swing (X and Y)  
choose 5 of the 15 at random; P(valid lineup)?

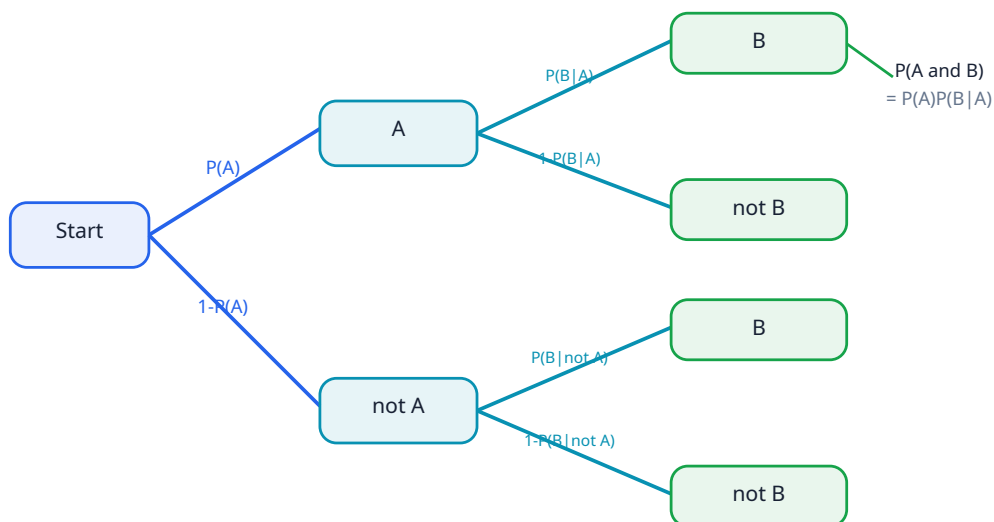
|                 |                                                   |
|-----------------|---------------------------------------------------|
| neither X nor Y | $C(5,2)C(5,2)C(3,1) = 300$                        |
| X only          | $C(5,1)C(5,2)C(3,1) \times 2 \text{ cases} = 300$ |
| Y only          | same = 300                                        |
| both X and Y    | $C(5,1)C(5,1)C(3,1) + 2C(5,2)C(3,1) = 135$        |
| TOTAL = 1035    |                                                   |

$$P = 1035 / C(15,5) = 1035/3003$$

THE METHOD: split into disjoint cases by WHO of the swing players is used, count each case, ADD. Never double count.  
no X, no Y + X only + Y only + both X and Y = total

## 12. Conditional probability (p050)

Conditional probability tree



Conditional probability tree

$$P(A|B) = P(A \cap B) / P(B), \quad P(B) > 0$$

Worked: digital camera buyers

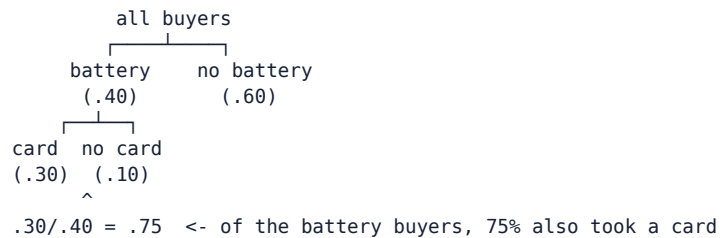
|                          |                     |
|--------------------------|---------------------|
| 60% buy a memory card    | $P(A) = .60$        |
| 40% buy an extra battery | $P(B) = .40$        |
| 30% buy both             | $P(A \cap B) = .30$ |

$$P(\text{card} \mid \text{battery}) = .30/.40 = .75$$

$$P(\text{battery} \mid \text{card}) = .30/.60 = .50$$

NOTE THE TWO ANSWERS ARE DIFFERENT. .75 is NOT .50.  
 The denominator is WHAT YOU ARE CONDITIONING ON.  
 $P(A|B)$  and  $P(B|A)$  are different numbers.

The tree that makes it obvious:



That is lecture 2 in full. The two things to carry forward: (1) the 36-cell dice grid, (2) exclusive is not independent.